

NLP BASICS

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- *Applications*
- *NLP Approaches*
- *NLP Components*
- *Phases in NLP*
- *Ambiguities*
- *NLP Preprocessing techniques*
- *POS Tagging*
- *Regular Expressions*

The Human Language



LANGUAGE



ALPHABETS

字母 वर्णमाला

ALFABETOS

الأبجدية எழுத்துக்கள்

The Human Language



LANGUAGE



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Words form Sentences

The Human Language

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Words form Sentences

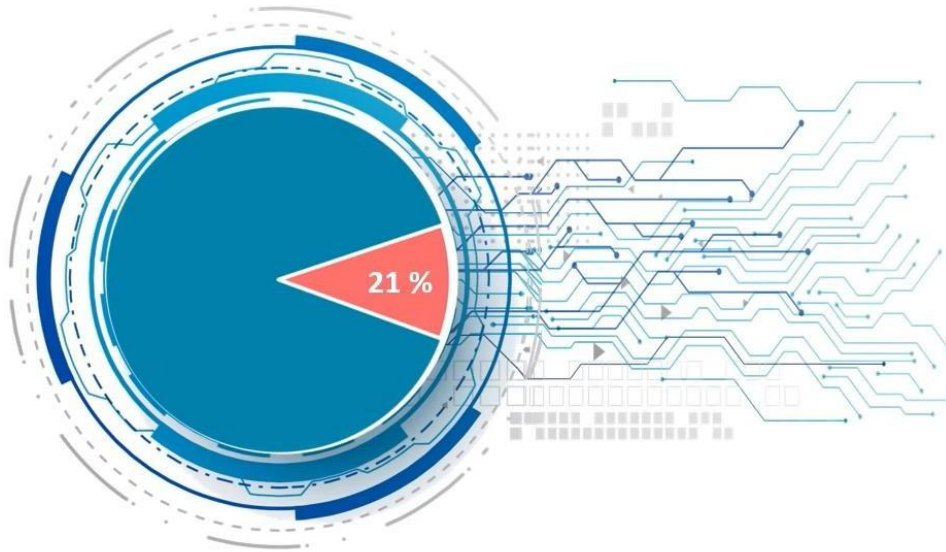
**KNOW THE
RULES !**



A language is not just words. It's a culture, a tradition, a unification of a community, a whole history that creates what a community is. It's all embodied in a language.

Noam Chomsky

The 21st Century



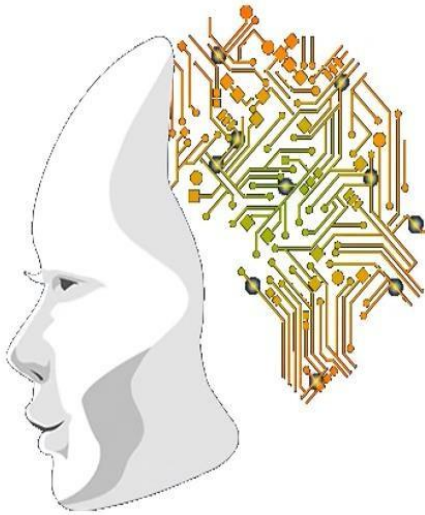
STRUCTURED
DATA



Text Mining / Text Analytics is the process of deriving meaningful information from natural language text

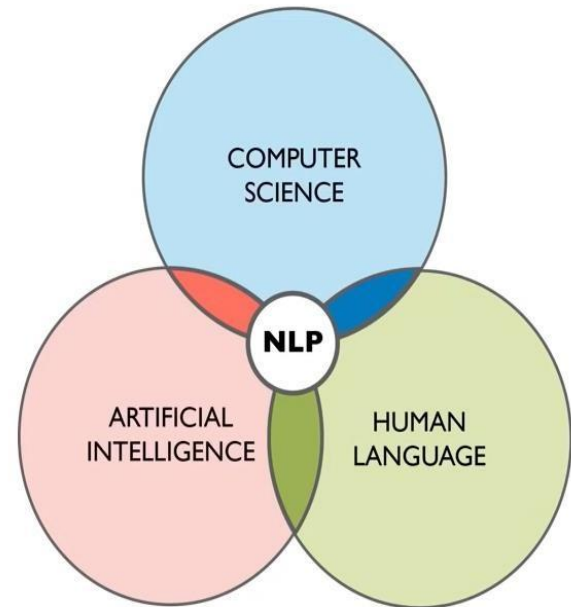


Text Mining and NLP



As, Text Mining refers to the process of deriving high quality information from the text .

The overall goal is, essentially to turn text into data for analysis, via application of Natural Language Processing (NLP)



Applications of NLP



**Sentimental
Analysis**

Chatbot



**Speech
Recognition**

**Machine
Translation**



Applications of NLP and Text Mining



Spell
Checking

Information
Extraction



Keyword
Search

Advertisement
Matching



Core Tasks



Text
Classification



Information
Extraction



Conversational
Agent



Information
Retrieval



Question
Answering Systems

General Applications



Spam
Classification



Calendar Event
Extraction



Personal
Assistants



Search
Engines

JEOPARDY!

Jeopardy!

Industry Specific



Social Media
Analysis



Retail Catalog
Extraction



Health Records
Analysis



Financial
Analysis



Legal Entity
Extraction

Main approaches in NLP

1. Rule-based Approach:

- This approach relies on predefined linguistic rules and patterns to process text.
- Linguistic experts and programmers manually create rules that encode knowledge about language.
- These rules are used to perform tasks such as tokenization, part-of-speech tagging, parsing, and information extraction.
- Rule-based systems are based on explicit, handcrafted rules and are effective for domains with well-defined rules.

2. Statistical Approach:

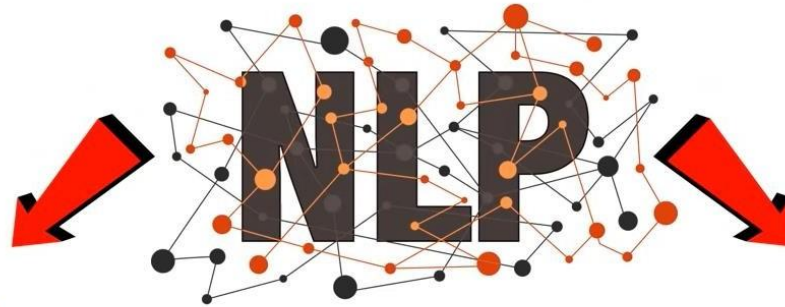
- Statistical NLP, also known as data-driven or machine learning-based NLP, utilizes statistical models and algorithms to learn patterns and structures from large amounts of annotated text data.
- These models use probabilistic techniques to make predictions about linguistic phenomena based on observed patterns in the training data.
- Statistical NLP techniques include machine translation, named entity recognition, sentiment analysis, and text classification.
- Statistical models require large amounts of annotated training data and can automatically extract relevant features from the data.

3. Neural Network Approach:

- Neural networks have revolutionized NLP in recent years.
- Deep learning models, such as recurrent neural networks (RNNs) and transformers, have shown remarkable performance in various NLP tasks.
- These models can learn hierarchical representations of text data and capture complex linguistic patterns.
- They excel in tasks such as language modeling, machine translation, sentiment analysis, and question answering.
- Neural network approaches require substantial computational resources and large amounts of annotated data for training.
- Distributional Semantics and Word Embeddings: Word embeddings, such as Word2Vec and GloVe, are vector representations of words that capture their semantic relationships and contextual information.

- **Pre-trained Language Models:** Pre-trained language models, such as BERT (Bidirectional Encoder Representations from Transformers) and GPT (Generative Pre-trained Transformer), have gained popularity in recent years.
- These models are trained on large-scale text data and capture rich linguistic representations.
- They can be fine-tuned for specific NLP tasks, requiring less task-specific data for training and achieving state-of-the-art performance in various tasks.

NLP : Components



Natural Language Understanding

Natural Language Generation

NLP : Components



**Natural Language
Understanding**

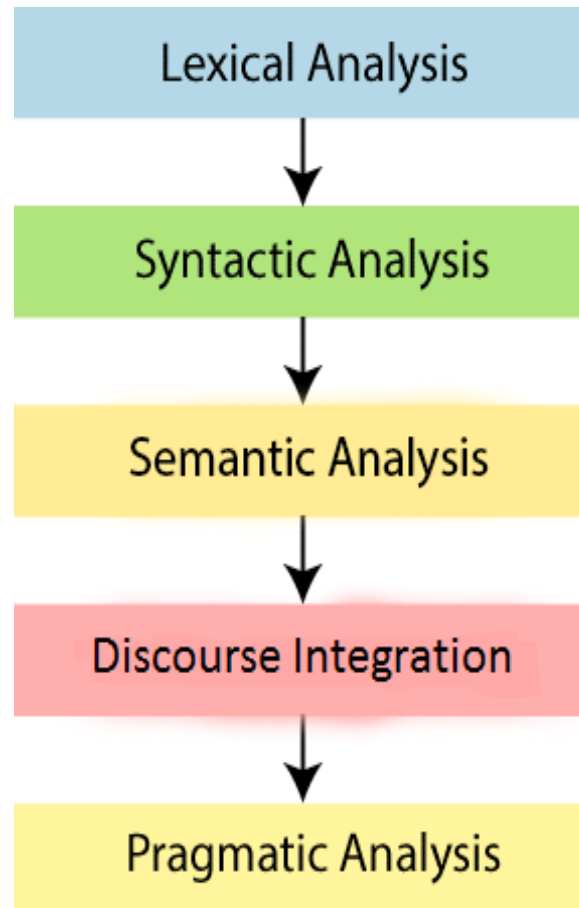
- Mapping input to useful representations
- Analyzing different aspects of the language



**Natural Language
Generation**

- Text Planning
- Sentence Planning
- Text Realization

Phases of NLP



Morphological Analysis/Lexical Analysis

- This phase scans the source code as a **stream of characters** and converts it into meaningful **lexemes**.
- It divides the whole text into **paragraphs, sentences, and words**.
- For example, irrationally can be broken into ir (prefix), rational (root) and -ly (suffix).
- It also assigns the possible Part-Of-Speech (**POS**) to the word.
- For example, the word “character” can be used as a noun or a verb.

The Two relationships

Two different kind of relationship among words

Inflectional morphology

Grammatical: number, tense, case, gender

Creates new forms of the same word: *bring, brought, brings, bringing*

Derivational morphology

Creates new words by changing part-of-speech: *logic, logical, illogical, logically, logician, logicize.*

Morphological Analysis

Input	Morphological Parsed Output
cats	cat +N +PL
cat	cat +N +SG
cities	city +N +PL
geese	goose +N +PL
goose	(goose +N +SG) or (goose +V)
gooses	goose +V +3SG
merging	merge +V +PRES-PART
caught	(catch +V +PAST-PART) or (catch +V +PAST)

Goal

To take input forms like those in the first column and produce output forms like those in the second column.

Output contains stem and additional information; +N for noun, +S G for singular, +PL for plural, +V for verb etc.

Syntactic Analysis (Parsing)

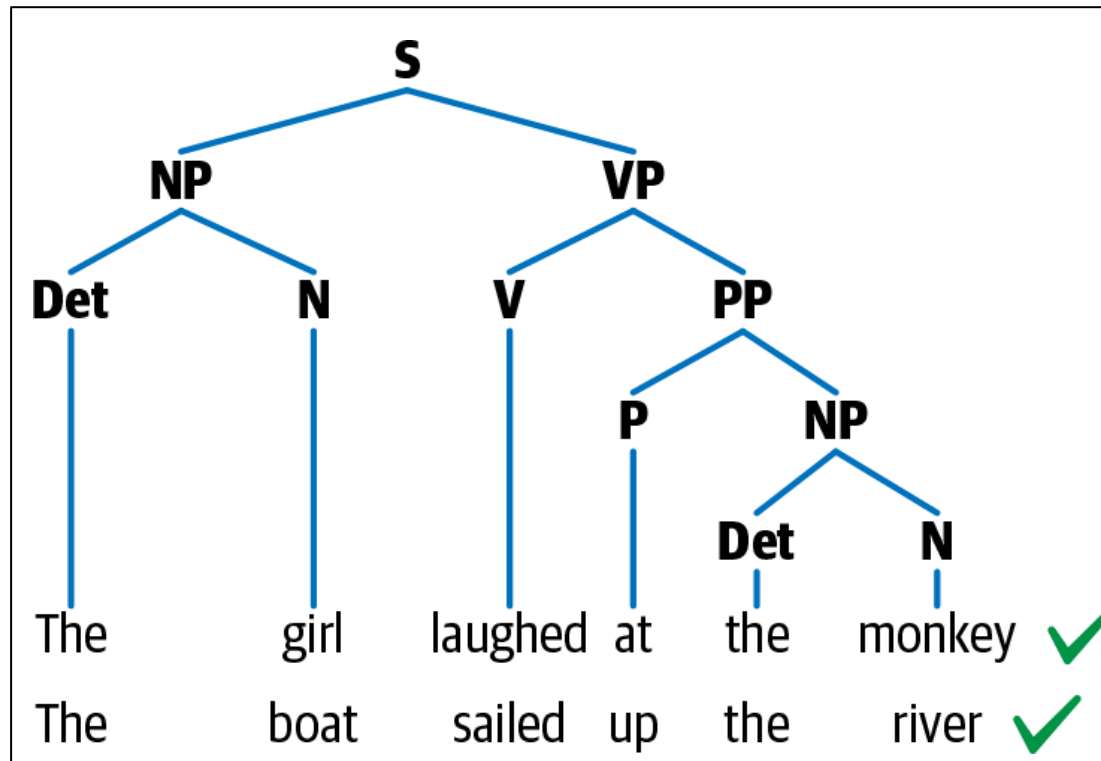
- Syntax is a set of rules to construct grammatically correct sentences out of words and phrases in a language.
- Syntax Analysis ensures that a given piece of text is correct structure.
- It tries to parse the sentence to check correct grammar at the sentence level.
- It also assigns the possible Part-Of-Speech (POS) to the word.
- For example:

Correct Syntax: Sun rises in the east.

Incorrect Syntax: Rise in sun the east.

Syntactic Analysis (Parsing)

- Syntactic structure in linguistics is represented in many different ways. A common approach to representing sentences is a parse tree.



Syntactic structure of two syntactically similar sentences

Semantic Analysis

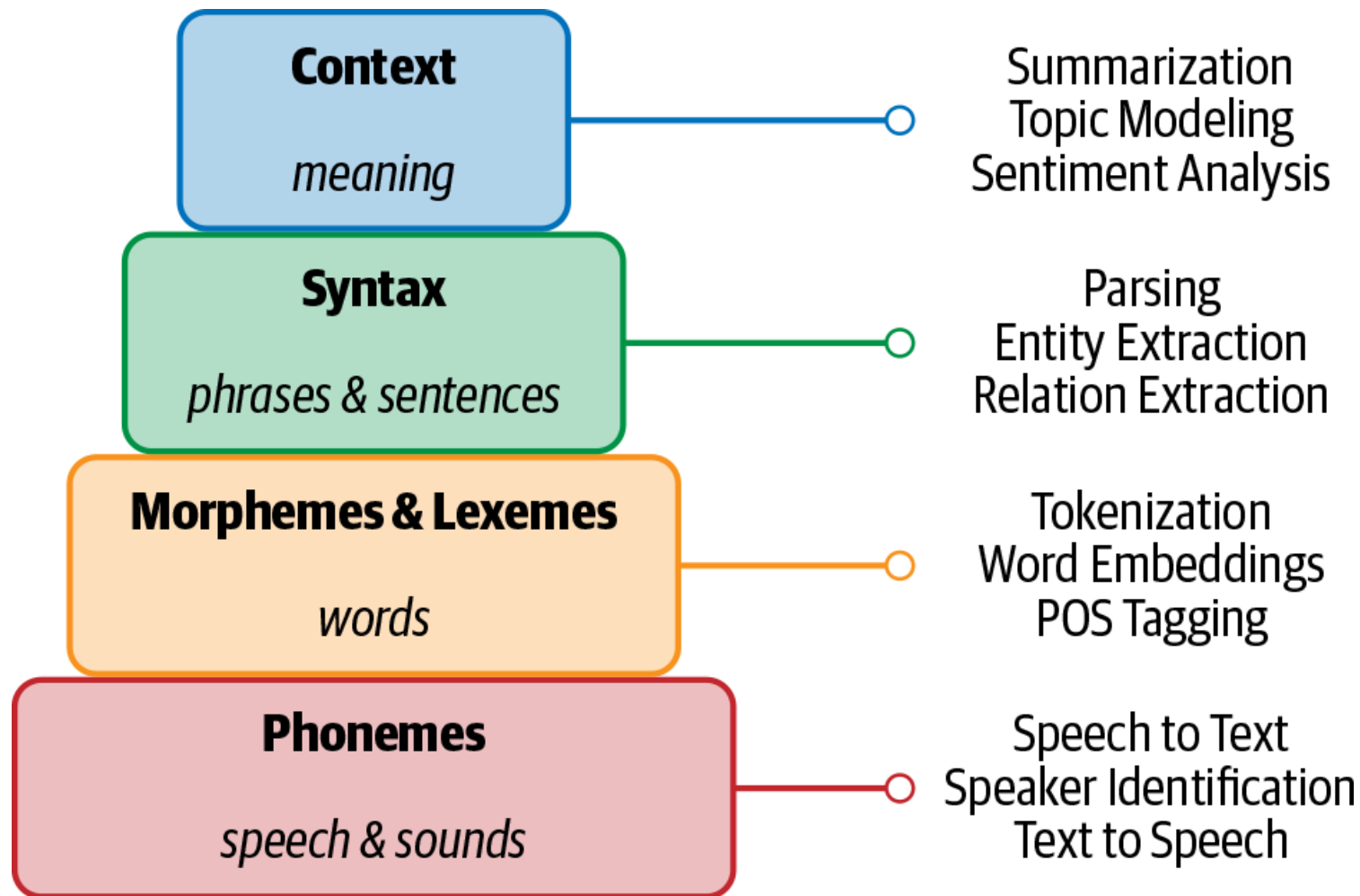
- Semantic analysis is concerned with the **meaning representation**.
- It mainly focuses on the literal meaning of words, phrases, and sentences.
- Consider the sentence: “The apple ate a banana”. Although the sentence is syntactically correct.
- It doesn’t make sense because apples can’t eat.
- Knowledge of wordnet

Discourse Integration

- Discourse deals with the effect of a previous sentence on the sentence in consideration.
- In the text, “Jack is a bright student. He spends most of the time in the library.”
- Here, discourse assigns “he” to refer to “Jack”.

Pragmatic Analysis

- Pragmatic is the fifth and last phase of NLP.
- It helps you to discover the intended effect by applying a set of rules that characterize cooperative dialogues.
- For Example: "Open the door" is interpreted as a request instead of an order.



Blocks of Language

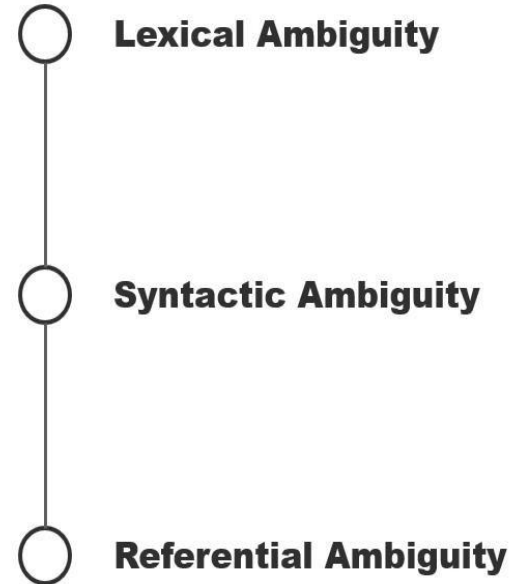
Applications

NLP : Ambiguity



**Natural Language
Understanding**

Ambiguity



NLU : Ambiguity

Ambiguity



Lexical Ambiguity

She is looking for a *match*.

The fisherman went to the *bank*.



Syntactic Ambiguity



Referential Ambiguity

NLU : Ambiguity

Ambiguity



Lexical Ambiguity

She is looking for a *match*.

The fisherman went to the *bank*.



Syntactic Ambiguity

The chicken is ready to eat.

Visiting relatives can be boring.

I saw the man with the binoculars.



Referential Ambiguity

NLU : Ambiguity

Ambiguity



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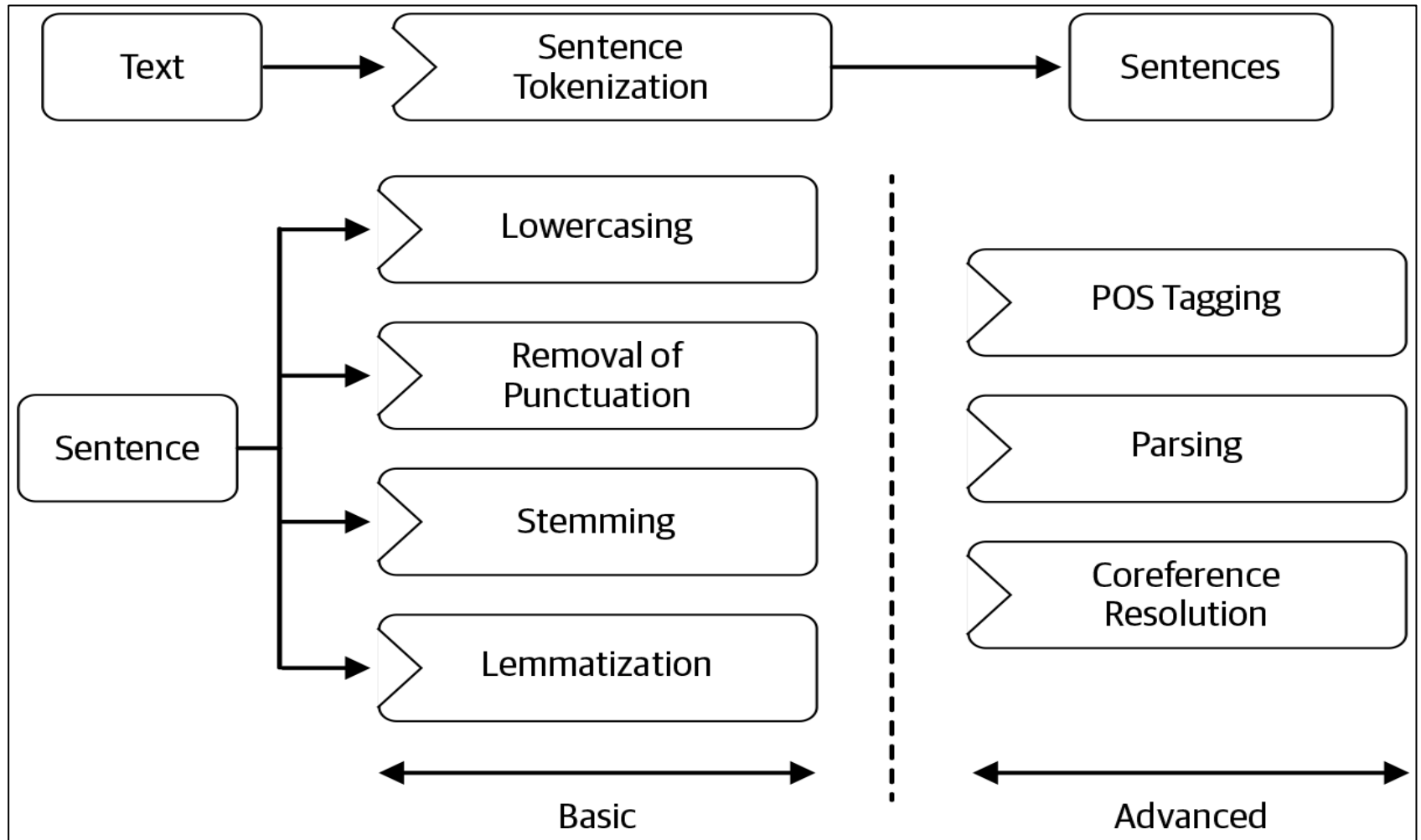


Referential Ambiguity

The boy told his father the theft. *He* was very upset

NLP Pre-Processing Techniques

Pre-Processing Steps



Tokenization

Tokenization is the first step in NLP

01

Break a complex sentence into words



02

Understand the importance of each of the words with respect to the sentence



03

Produce a structural description on an input sentence



Tokenization

Tokenization

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01

Break a complex sentence into words



02

Understand the importance of each of the words with respect to the sentence



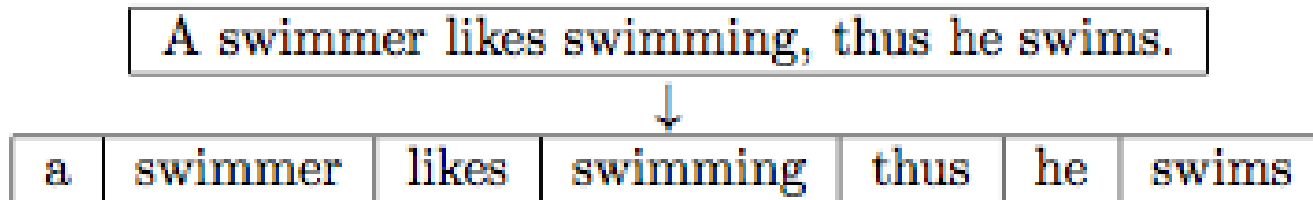
03

Produce a structural description on an input sentence



Word Tokenization

Tokenization is the process of segmenting a string of characters into tokens (words).



An example

I have a can opener; but I can't open these cans.

- Word Tokens: 11
- Word Types: 10

Several tokenization libraries

- NLTK Toolkit (Python)
- Spacy (Python)
- Polyglot (Python)
- Stanford CoreNLP (Java)
- Unix Commands

Stemming

Normalize words into its base form or root form



Affectation

Affects

Affections

Affected

Affection

Affecting

Stemming

Normalize words into its base form or root form

Affect



Lemmatization example

WordNet lemmatizer

- Uses the WordNet Database to lookup lemmas
- `nltk.stem.WordNetLemmatizer`
- Examples:

– feet → foot

cats → cat

– wolves → wolf

talked → talked

- Problems: not all forms are reduced
- Takeaway: we need to try stemming or lemmatization and choose best for our task



What are Named Entity Recognition ?



MOVIE



MONETARY VALUE



ORGANIZATION



LOCATION

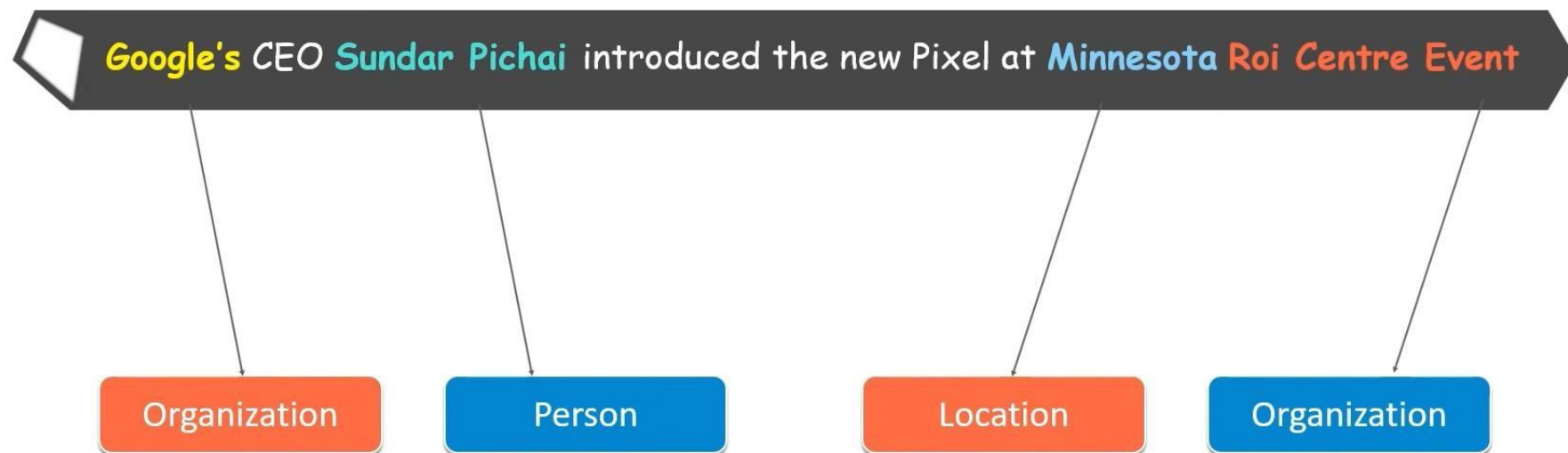


QUANTITIES

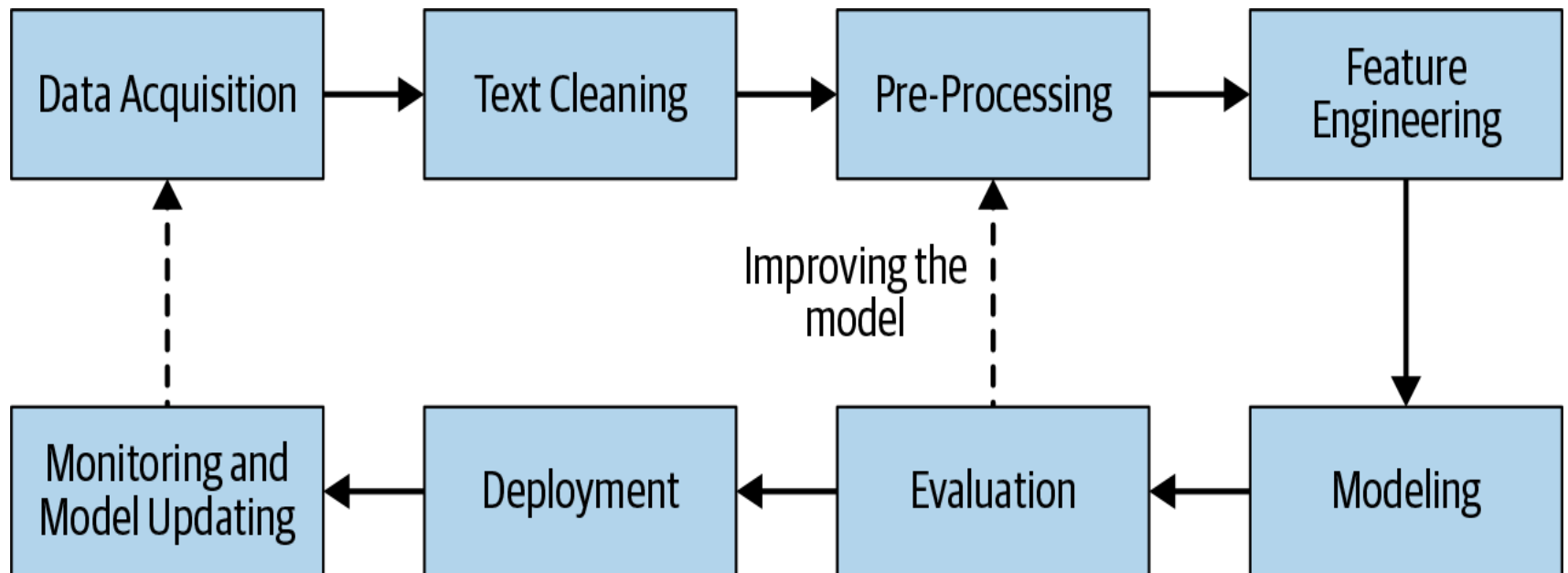


PERSON

NER : Named Entity Recognition



Generic NLP Pipeline



References:

- Speech and Language Processing (3rd ed. draft) by Dan Jurafsky and James H. Martin
<https://web.stanford.edu/~jurafsky/slp3/>
- Practical Natural Language Processing (O'Reilly)
A Comprehensive Guide to Building Real-World NLP Systems
By Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, and Harshit Surana

Thank you